

Harshana Lakshara Fernando

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Summary

Software Engineer with production experience building Java/Spring Boot microservices, Kafka-backed distributed systems, and gRPC integrations. Delivered backend services and rollout strategies for production systems, including a Kafka-backed PNR generation service handling 25+ PNRs/sec at <100ms latency, with additional experience in PostgreSQL, Go/Gin, Docker, Kubernetes, CI/CD, and observability.

Skills

Core	Java Spring Boot Microservices Distributed Systems System Design REST APIs
Backend	Spring Cloud Spring Security Resilience4j Node.js Express.js Gin
Messaging & APIs	Kafka RabbitMQ gRPC GraphQL WebSockets Protobuf JWT OAuth 2.0
Data	PostgreSQL MySQL Oracle DB Redis Couchbase
Cloud & DevOps	AWS Docker Kubernetes CI/CD (GitHub Actions, Jenkins)
Testing	JMeter Gatling Pytest Unit Testing
Tools	Git Gradle Postman Jira Confluence

Experience

R&D Software Engineer — KōreroCare Platform

Mar 2026 – Present

Empathic Computing Laboratory, University of Auckland

Remote

- Co-led backend design for the **KōreroCare** mobile-wearable sensing platform, building **Go/Gin**-based services for time-series ingestion and cross-device synchronization.
- Designed **PostgreSQL** schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated **Keycloak OAuth2/OIDC** with **Traefik** gateway routing and Docker Compose deployment for secure internal service access.
- Developed Firebase notification and reminder flows for participant engagement and study prompts.
- Modeled an **HL7/FHIR**-aligned data schema mapping participants, devices, and wearable measurements.

Software Engineer Intern

Dec 2024 – May 2025

Information Systems Associates (Pvt) Ltd — Aviation Technology (AirArabia)

- AeroOrder — Order Management Microservice (Java 17, Spring Boot, Gradle)**
 - Built a **distributed Kafka-based PNR ID generator** (live in **production**), ensuring **uniqueness and consistency** across horizontally scaled services.
 - Achieved **25+ PNRs/sec per node** with **<100ms latency**, validated using **Apache JMeter** (load, stress, spike testing).
 - Improved **reliability and recoverability** by resolving **Kafka consumer lag and offset issues** using **graceful shutdown hooks** and **retry mechanisms**.
 - Implemented **observability (Prometheus, Grafana)** for real-time monitoring of **queue time and latency**.
 - Refactored **Kafka Streams pipelines** into a **Producer-Consumer model**, reducing lag and improving stability.
 - Improved **maintainability and debugging** via **structured JSON logging** and removal of legacy dependencies.
 - Maintained **>90% unit test coverage** while delivering **production-ready features**.
- AeroMart — Legacy Passenger Reservation System (Java 8, Ant)**
 - Integrated **gRPC clients** and **Kafka consumers** into a **Java 8 monolith**, enabling **backward-compatible communication** with microservices.
 - Implemented **database-driven feature toggles** for **PNR routing**, enabling **safe, phased rollouts with fallback**.

• Engineering Practices

- Authored **design documents, sequence diagrams, and integration plans**; contributed to **Agile workflows, code reviews**, and team collaboration (Jira, Bitbucket, Confluence).

KEYWORDS: Java 17/8, Spring Boot, Kafka, Kafka Streams, gRPC, Docker, Prometheus, Grafana, JMeter, Oracle DB, Microservices, Distributed Systems, Time-Series Data.

Freelance Software Engineer — Selected Client Projects

May 2025 – May 2026

Client Platforms

Remote / Sri Lanka

- Delivered **AgriNova**, a multi-branch inventory and sales platform, owning backend workflows, deployment, and client-facing fixes end-to-end.
- Implemented transaction-safe stock logic across **3 core workflows**: material issuance, returns, and purchase operations.
- Resolved customer-reported production issues including authentication/version compatibility problems and workflow-level validation bugs.
- Reduced repeat dashboard data retrieval latency from **~384ms to <1ms** by implementing Next.js/React Query caching, pagination, and **PostgreSQL** indexing.
- Set up deployment and CI/CD workflows for full-stack delivery using Vercel and Supabase/PostgreSQL.

Projects

See more projects on my portfolio [↗](#)

1. Employee Management System (EMS) (2025 | Individual)

Architecture | Github 

Designed and built a comprehensive, cloud-native microservices application for employee and department management, featuring a full-stack architecture with a React/TypeScript frontend and Spring Boot backend.

- Architected a distributed system using **Spring Cloud** for service discovery, centralized configuration, and an API gateway, with **Resilience4j** for robust fault tolerance.
- Enabled dynamic configuration management using **Spring Cloud Bus** with **RabbitMQ** and established a robust observability stack (Zipkin, ELK, Prometheus/Grafana).
- Designed and managed a CI/CD pipeline using **GitHub Actions**, with **ArgoCD** for GitOps-based deployments to the Kubernetes cluster.

Keywords: Java 17, Spring Boot, Spring Cloud, React, TypeScript, Resilience4j, RabbitMQ, Docker, Kubernetes, GitHub Actions, ArgoCD, Prometheus, Grafana

2. GoTogether (2024 | Group)

Architecture | Live  | Github 

Developed and deployed a full-stack travel assistant platform with a **Next.js** frontend (SSR/CSR, secure auth) and microservices in **Spring Boot** and **Go**, enabling AI itinerary generation via **Gemini API**.

- Implemented a **Spring Boot** social service with **Keycloak** integration for identity management and a dedicated **Go** service using **gRPC** for high-performance inter-service communication.
- Configured production-grade security with **NGINX**, **Certbot**, and **DuckDNS**, and orchestrated deployment on **Oracle Cloud Free Tier** using custom scripts and Docker optimizations.
- Built a dedicated **Auth Service using Keycloak**, managing OAuth2/OIDC flows, token issuance, and automated realm/client boot-time import scripts.

Keywords: Next.js, Go, Spring Boot, Keycloak, PostgreSQL, NGINX, Docker, Gemini API, Oracle Cloud, gRPC

3. Serverless Auction Platform (2025 | Individual)

Architecture | Github 

A capstone project built using AWS, following a microservices architecture with an event-driven, serverless (FaaS) design.

- Architected an event-driven system using **AWS Lambda**, **API Gateway**, and **EventBridge**, with **SQS** and **SNS** for asynchronous communication.
- Designed a high-performance **DynamoDB** data model with GSIs and implemented a custom **Lambda Authorizer** for JWT-based security.
- Utilized **LocalStack** and **Docker** for local cloud emulation and integrated **Middy** middleware for robust error handling.

Keywords: Node.js, Serverless Framework, AWS Lambda, DynamoDB, SQS, SNS, SES, JWT, Middy, LocalStack, Docker

4. Food Delivery Application (2024 | Group)

Github 

Built a **microservices-based, event-driven food delivery platform**.

- Developed **Restaurant service** with **JWT-based authentication** and CRUD APIs with **Redis** caching.
- Enabled **real-time order synchronization and notifications** using **Kafka** and **WebSockets**.

Keywords: Java, Spring Boot, gRPC, Kafka, Redis, WebSocket, Docker, Spring Security

5. Maison Vella – E-Commerce Admin Panel (2024 | Individual)

Architecture | Live

| Github

Comprehensive admin interface for the Maison Vella e-commerce ecosystem, facilitating centralized management of products, orders, and content assets.

- Engineered a responsive **Next.js 14** app with **Server Actions**, **tRPC**, and **ShadCN UI**, implementing **CQRS** and layered architecture.
- Integrated **Sanity** (Headless CMS) via GraphQL and managed data with **Prisma** and **Supabase** in a serverless environment.
- Secured access with **Clerk Auth** and optimized media delivery using **Cloudflare R2** and a global CDN.

Keywords: Next.js 14, Server Actions, tRPC, GraphQL, Prisma, Supabase, Sanity, Serverless, Cloudflare R2, CQRS

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering

2022 – 2026

University of Moratuwa, Sri Lanka

G.C.E. Advanced Level (Physical Science Stream)

2019

St. John’s College, Panadura, Sri Lanka

Results: 3 A’s (Combined Mathematics, Physics, Chemistry)

District Rank: 17

Professional Training, Coursework & Certificates

See all certificates

- Certified Kubernetes Administrator (CKA) Training**

CodeKloud (Mumshad Mannambeth) — Completed 27h curriculum with 50+ hands-on labs.

Apr 2025

View certificate
- AWS Certified Solutions Architect Associate (SAA-C03) Training**

Udemy (Stephane Maarek) — Completed 27.5h comprehensive cloud architecture course.

Aug 2025

View certificate
- Apache Kafka Series (v3) – Learn Kafka for Beginners**

Udemy (Stephane Maarek) — Full System Design & Setup.

Mar 2025

[Part 1] | [Part 2] | [Part 3]
- 100 Days of DevOps (Bootcamp)**

CodeKloud — Intensive hands-on training covering Linux, Python, Docker, Jenkins, Terraform & Ansible.

Nov 2025

View certificate
- Implementing Software Architecture for Large-Scale Systems**

Udemy

Apr 2025

View certificate
- Advanced OAuth Security**

Udemy

Mar 2024

View certificate
- Next.js: Build Scalable React Apps with Page & App Routers**

Udemy — Instructors: Anton Voroniuk, Denys Kohut, Anton Voroniuk Support

Nov 2025

View certificate

Honors & Open Source

- Dean’s List:** Semester 6; **Accepted Contributor:** GSSoC 2026 *AI & Agents Track*, SSOC Season 5.

Articles

See all articles

- The Architecture of Trust: The Ultimate Guide to API Security**

Layered API defense: AuthN vs. AuthZ, JWT/OAuth flows, and OWASP risks like BOLA.

API Security

Read article
- How Kafka Cheated the Two Generals’ Problem to Achieve Exactly-Once Semantics**

Distributed Systems

Deep technical breakdown of Kafka idempotence, transactions, and zombie fencing.

Distributed Systems

Read article

Referees

Dr. Kunal Gupta Research Fellow, Empathic Computing Lab University of Auckland Email: kunal.gupta@auckland.ac.nz	Dr. Dulani Meedeniya Professor, Dept. of Computer Science University of Moratuwa Email: dulanim@cse.mrt.ac.lk
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